

01 Introduction

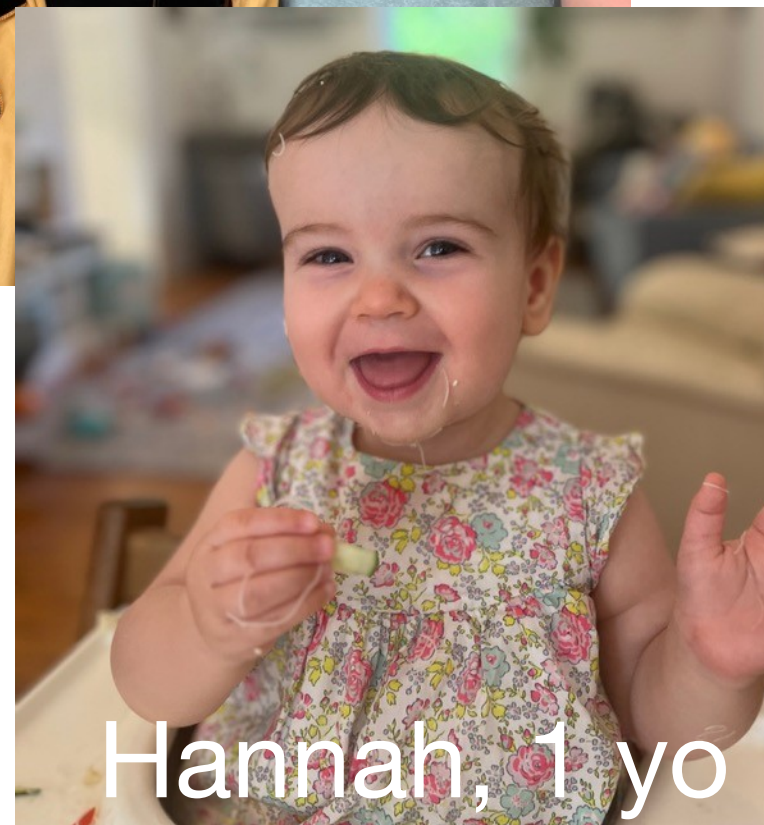
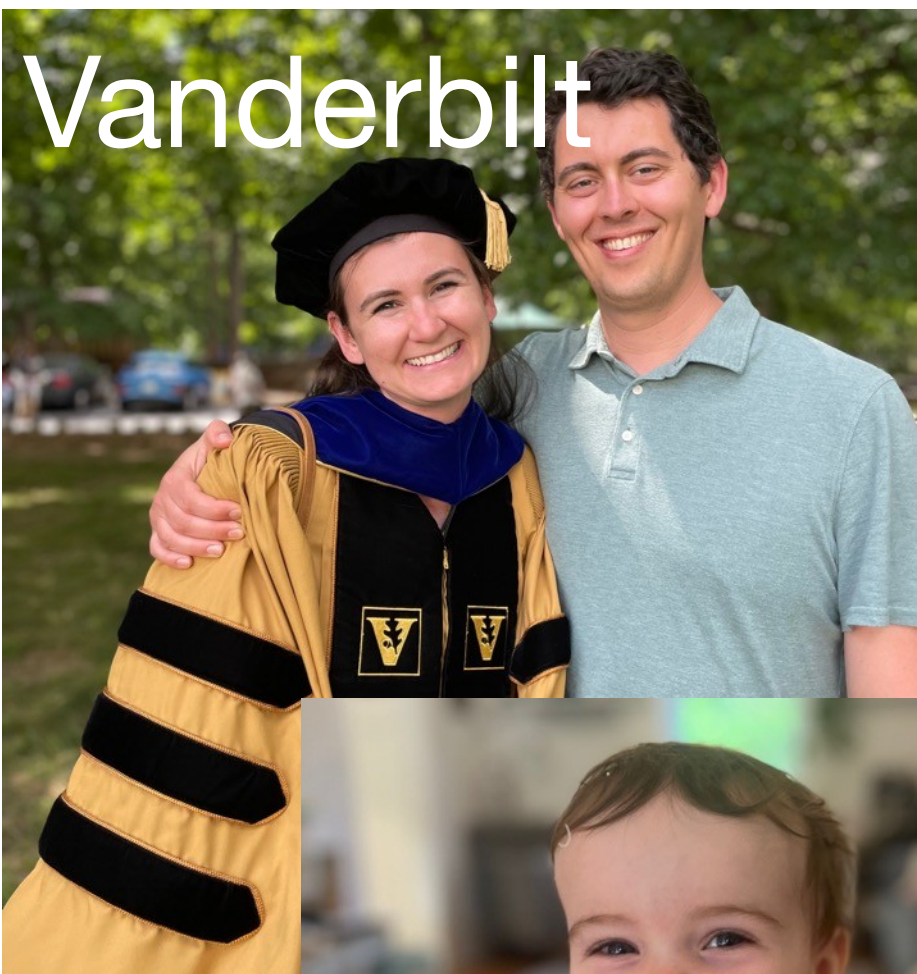
Sarah Groves

Welcome to DS2003!

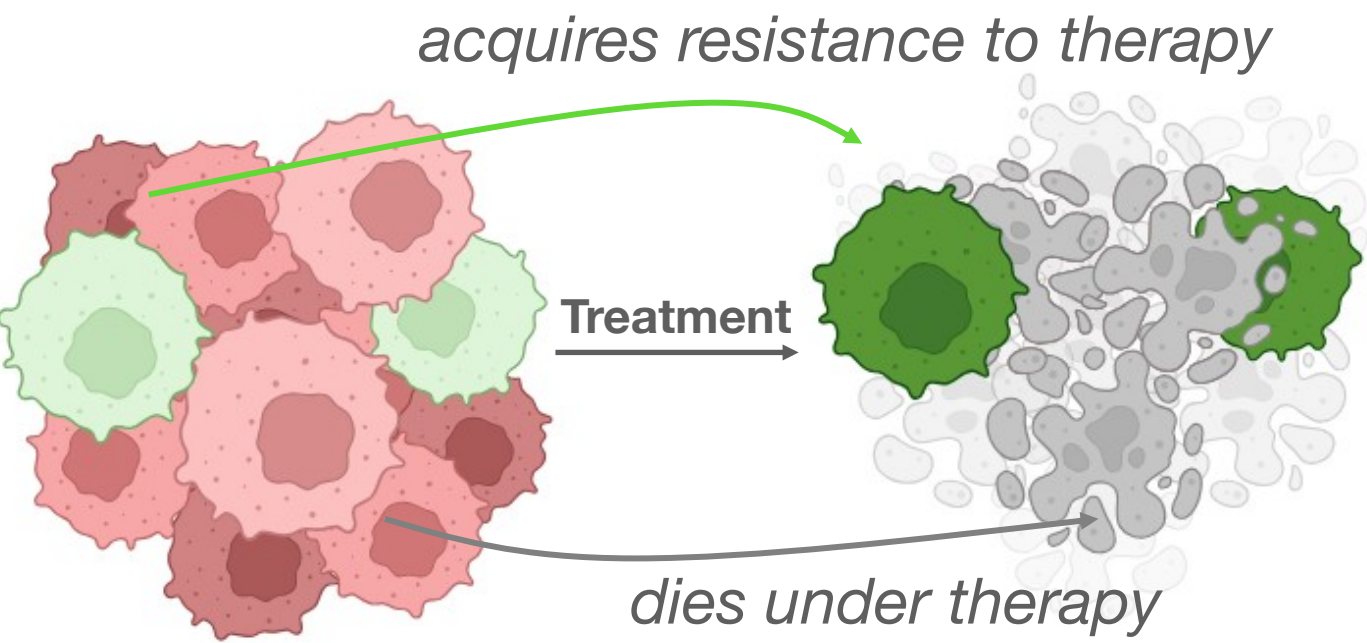
We are glad you are joining us 🎉

Who are we?

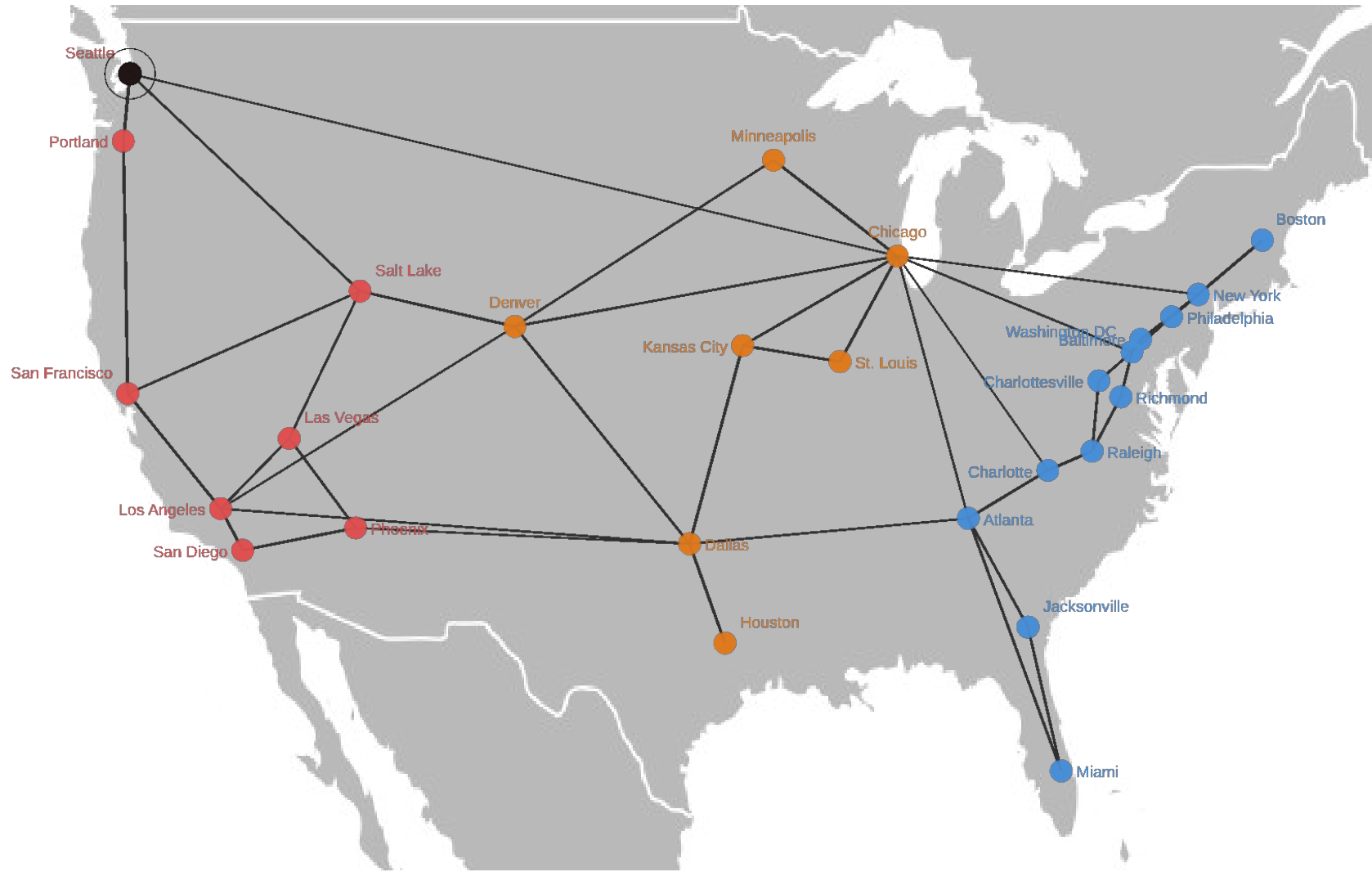
Sarah Groves



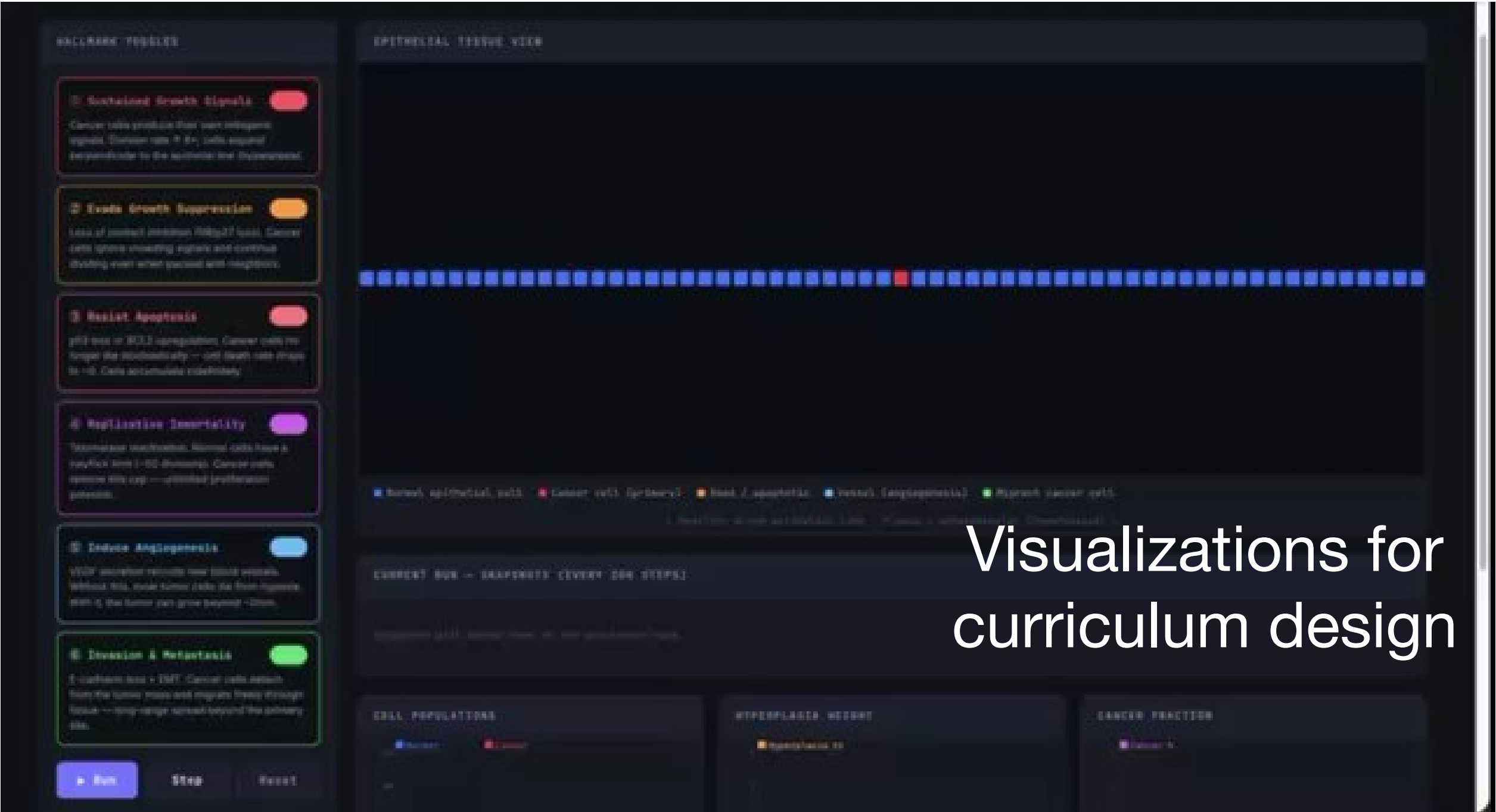
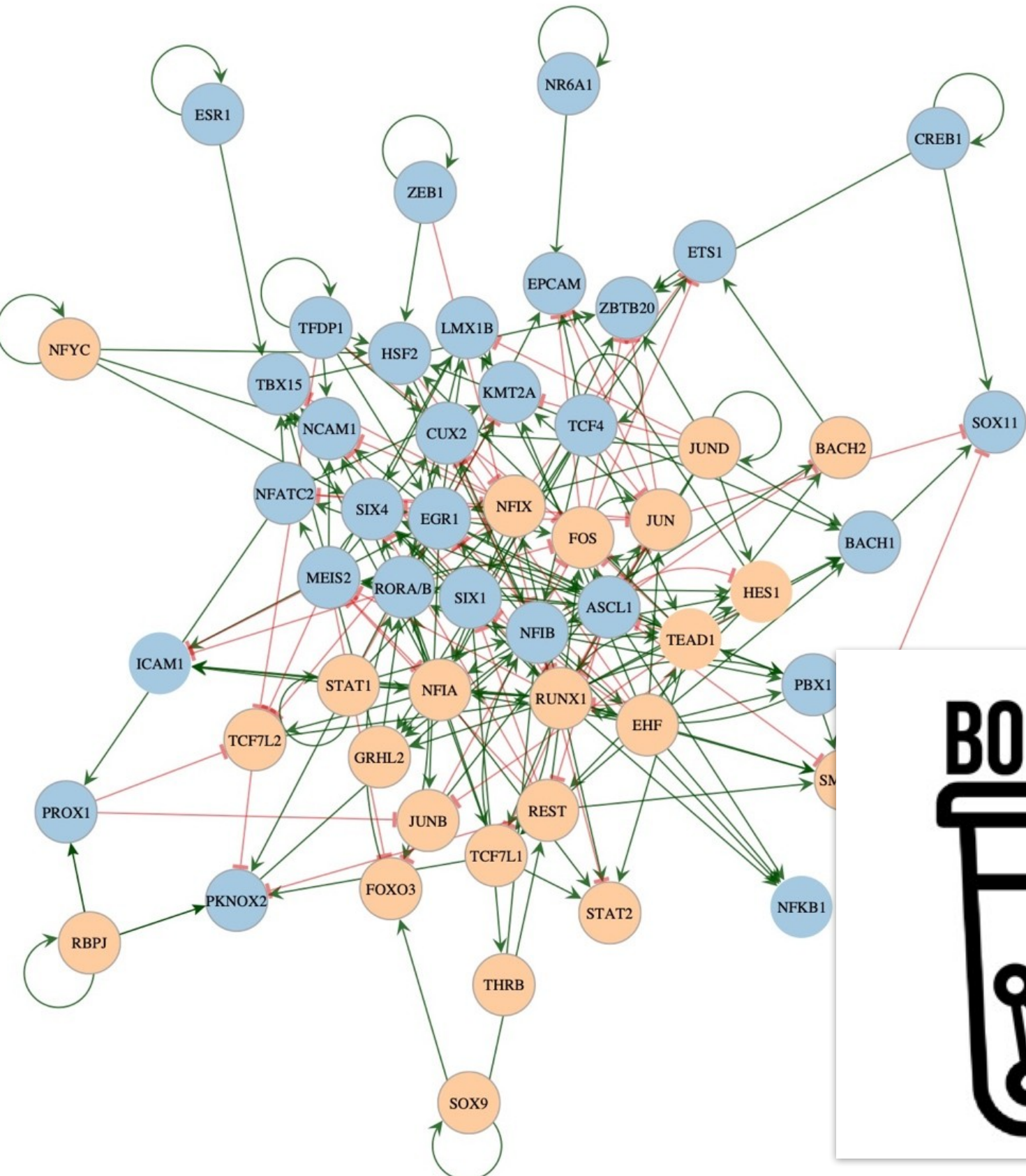
My research: biomedical network science



“NETWORKS,
NETWORKS
EVERYWHERE”



Transit
networks and
traffic



Visualizations for
curriculum design

TBD

DS2003 IA

Make a name card!

- Your name
- Major
- Research interests, if any
- At least 1 of your interests/hobbies

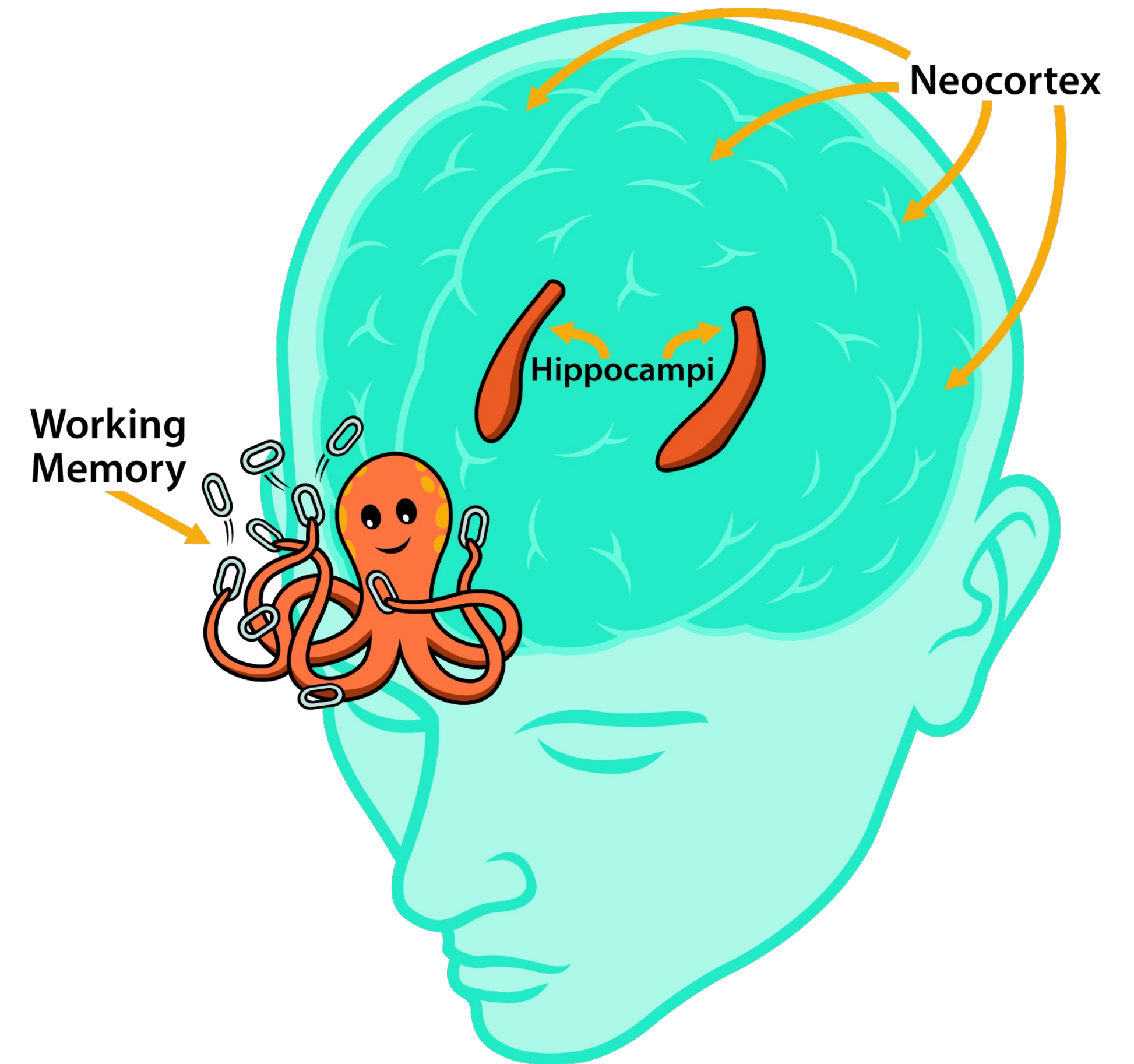
**What is
this class?**

Course Learning Objectives:

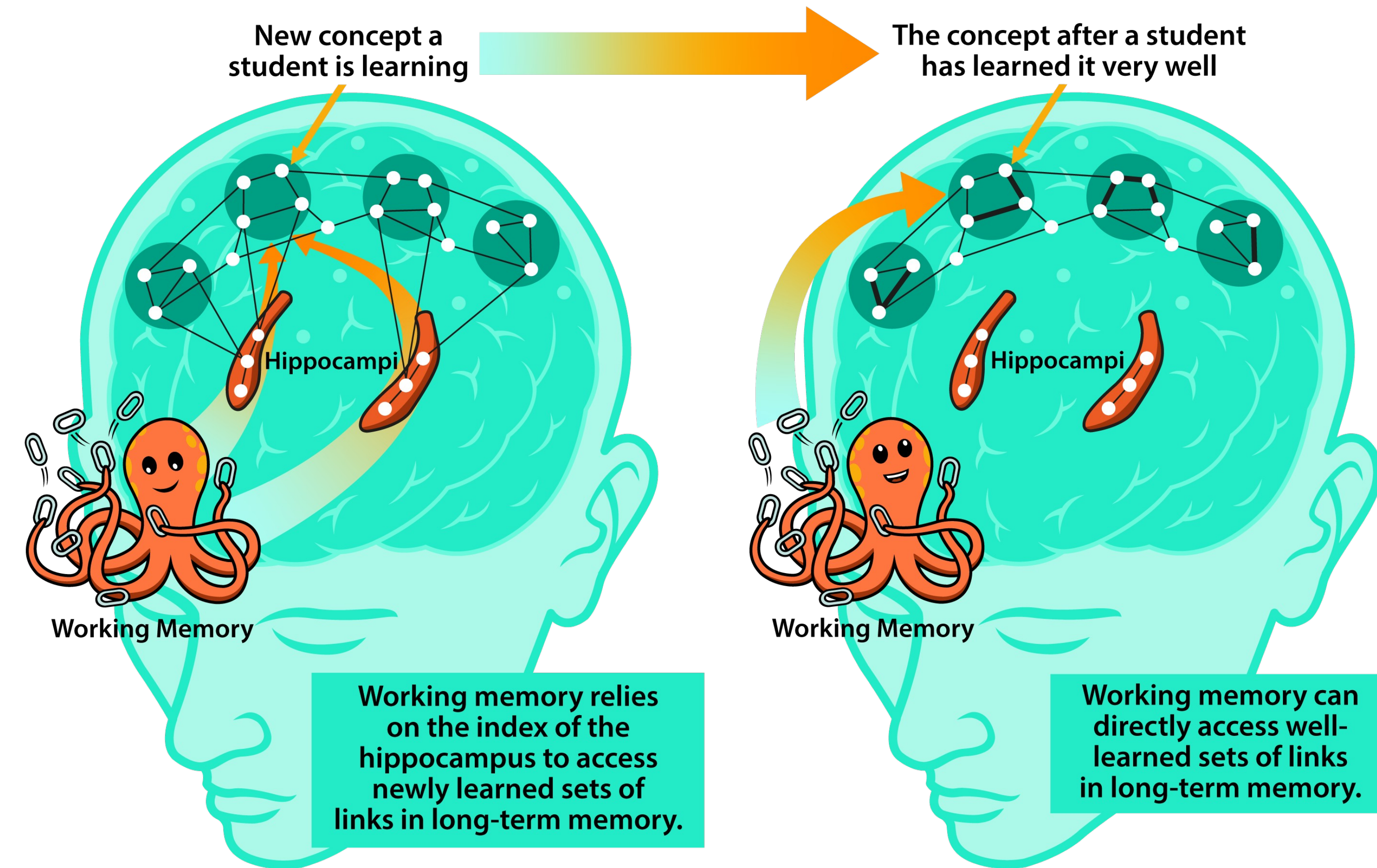
- Understand the importance of **data communication** in various domains.
 - Gain knowledge about various tools and techniques, including proficiency in using **industry-standard software for data visualization and reporting**.
 - Apply basic **design principles** to create visually effective and informative data visualizations.
 - Develop **storytelling** arches to richly describe data alongside visualizations.
 - Analyze how charts can **misrepresent data**.
 - Critique the **effectiveness** of various data communication strategies and make informed decisions about which techniques are best suited for specific communication objectives and target audiences.
-
- Can you **interpret** data visualizations critically?
 - Can you **produce** data visualizations that are effective?
 - For **exploring** data?
 - For **explaining** something?

Teaching philosophy (based on scientific studies):

- Students learn best when they feel like they belong and feel psychologically safe.
- We are all *unique learners* and we learn in our own ways.
- Courses should provide students with employable skills and experiences.
- Learning experiences should be authentic (“real world”).
- Active learning promotes skill-development and knowledge retention.
- Every student is capable of earning an “A”.
- Students can, and should, learn from their peers.
- Data scientists work better in teams.
- Data scientists-in-training should always be working on their communication and collaboration skills.



Learning how to learn, briefly



Building new links (synapses) is physically hard!

Let's make it more effective:

- Spaced retrieval practice
- Alternate focused and diffuse thinking
- Deliberate practice with feedback
- Visual, auditory, and hands-on modes of delivery
- Desirable difficulties

My expectations of each student:

- Show up
- *Engage*
- Trust that the course structure and assessments are intentional for your learning but ask me for my reasoning if it isn't clear!
- Ask questions and reach out for help when you need it
- Give me feedback about how your learning is going and how we can help
- Be patient with me as I teach this course for the first time

We have a social contract: I will teach to the best of my ability and you will learn to the best of your ability. Neither of us can do our part without the other

Student expectations of me & this class:

- Fairness in treatment and grading
- Availability
- Pacing of content
- Consistent communication

AI Policy: my expectations for you

- See the syllabus
- In general, you can use AI to help your learning, but don't use AI to replace your learning
 - AI convinces your working memory octopus part of your brain that you *know* something, without every encoding it in long term memory
 - Using AI to do things for you handicaps the procedural learning process severely
- If you do use AI, cite it

AI: your expectations for me

Privacy

Don't generate slides with AI

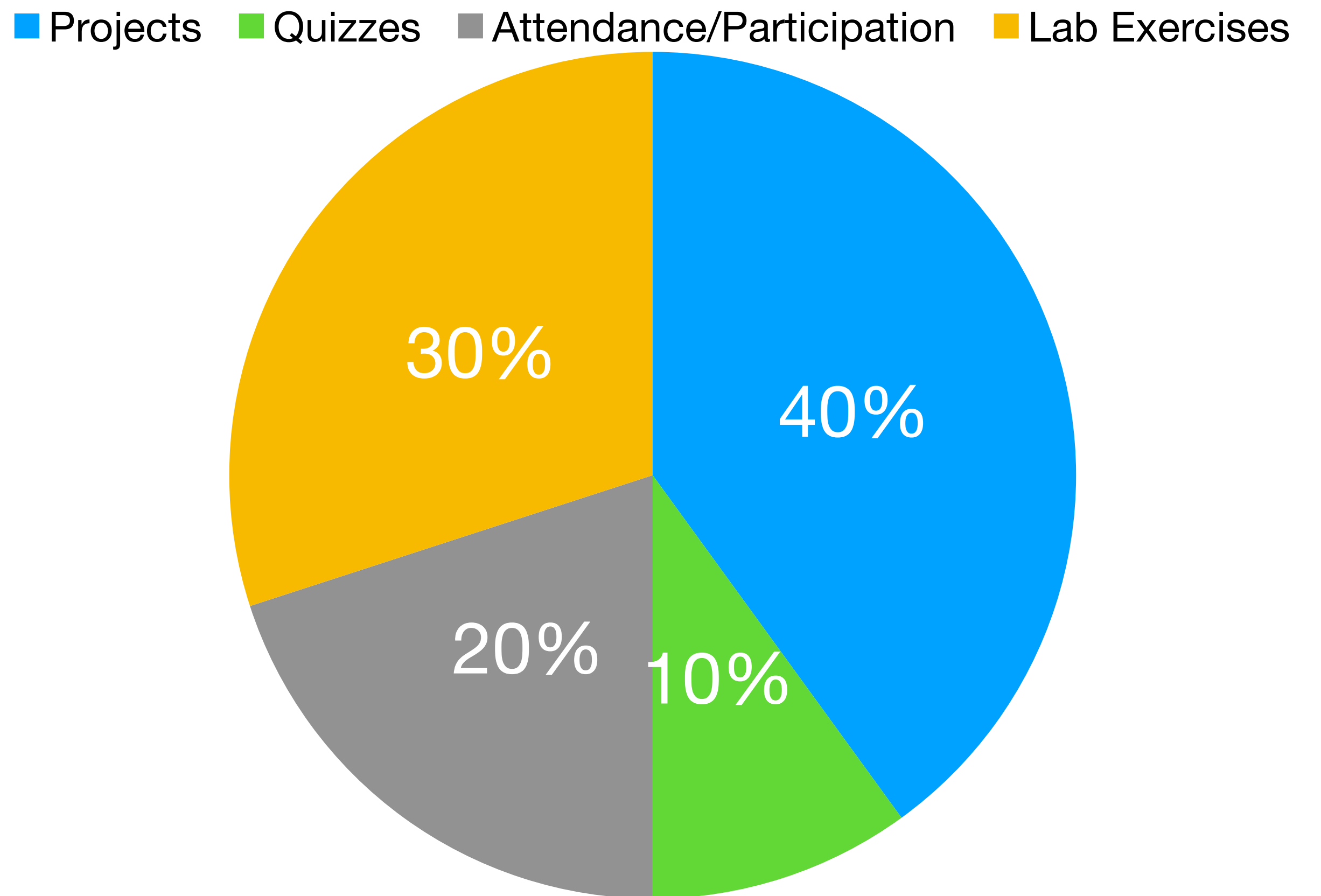
Class Schedule

- Lectures (Mon | Wed)
 - About the "what" and "why" **Multiple modes of delivery**
Focused and diffuse thinking
 - Review theory, research, visualization examples
 - **In-class activities** (e.g., critique good and bad visualization designs)
 - Wednesday: in class **quizzes**; drop lowest 2
- Labs (Fri) **Spaced retrieval practice**
 - About the "how," with coding in Python
 - I do, we do, you do **Desirable difficulties**
 - Weekly demos and in-class **lab** exercises, finish for homework & submitted by Sunday @ midnight **Deliberate practice with feedback**

Grading

- 30%: Labs
- 10%: In-class quizzes, drop lowest 2
- 20%: Attendance/participation
- 20%: Project 1: Exploratory data analysis
- 20%: Project 2: Interactive Data App, SciVis, or infographic

If you can't make class,
please email me as
soon as possible
(preferably before
class!)



By next class...

- Download **anaconda** (comes with its own version of Python; advanced: any Python)
- Download **VS code** (advanced: any IDE you want)
- Fill out the pre-course **survey** on CANVAS

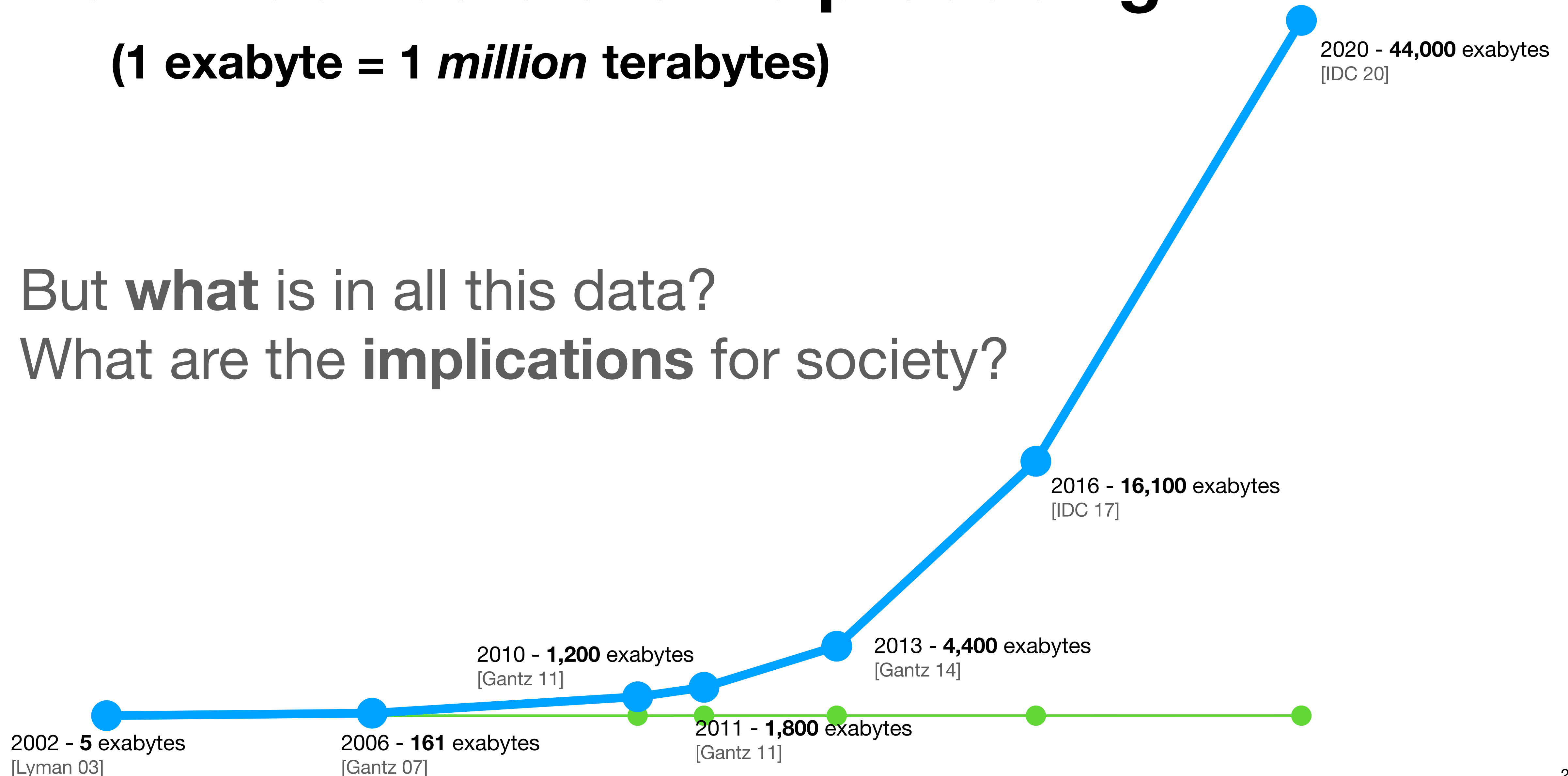
**What is
data visualization?**

How much data are we producing?

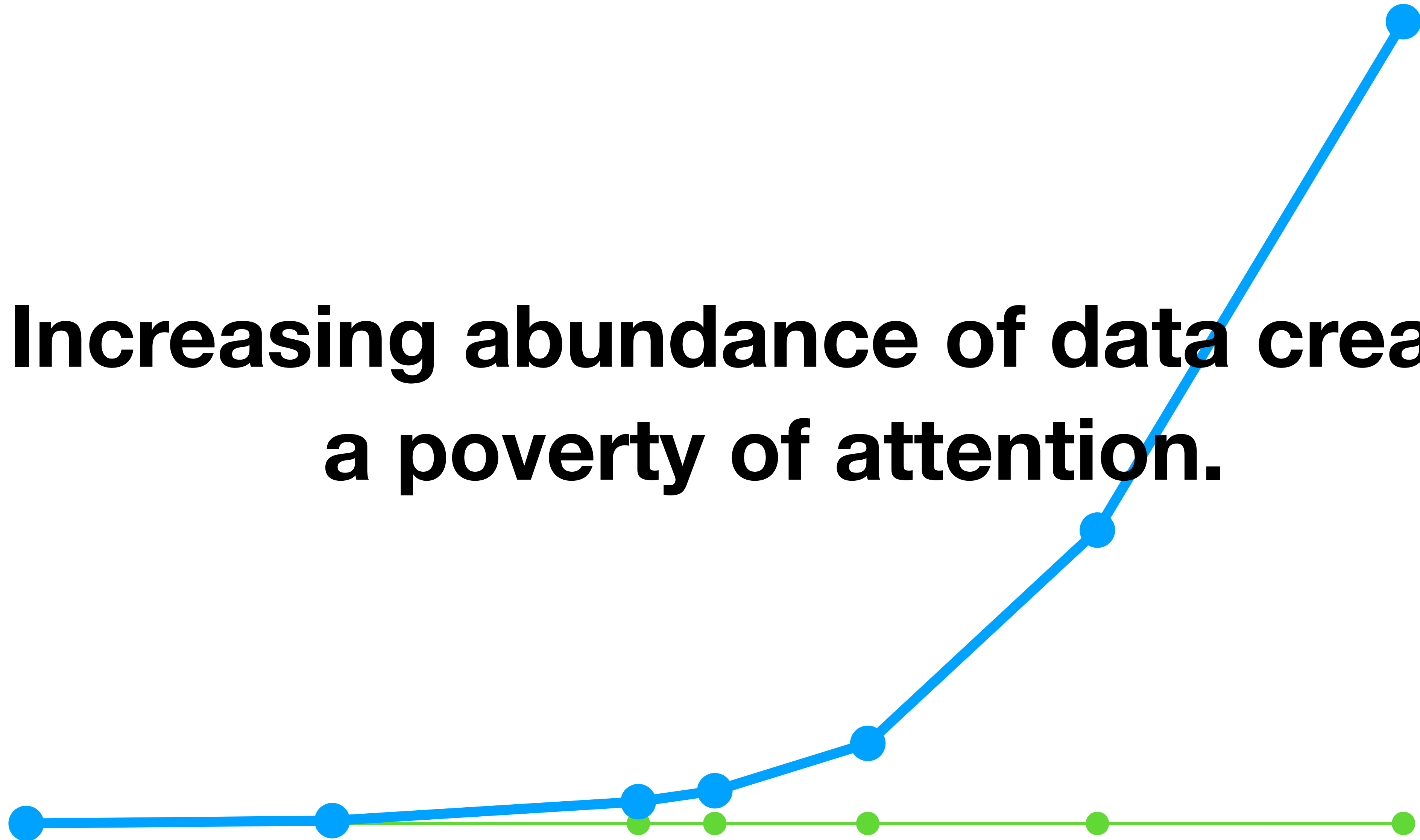
(1 exabyte = 1 *million* terabytes)

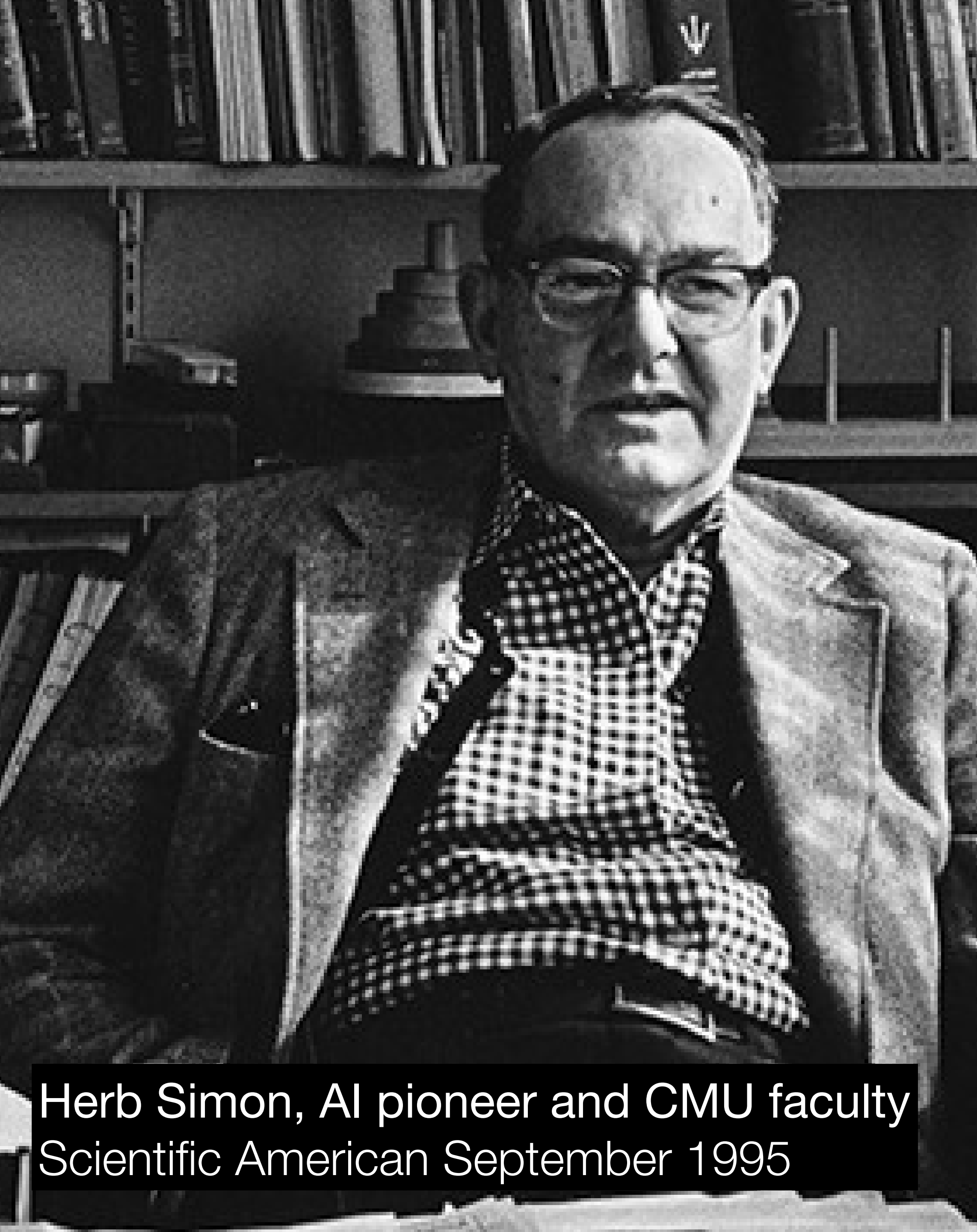
But **what** is in all this data?

What are the **implications** for society?



**Increasing abundance of data creates
a poverty of attention.**





A poverty of attention

What information consumes is rather obvious: it **consumes the attention** of its recipients. Hence a wealth of information creates a poverty of attention, and a need to **allocate that attention efficiently** among the overabundance of information sources that might consume it.

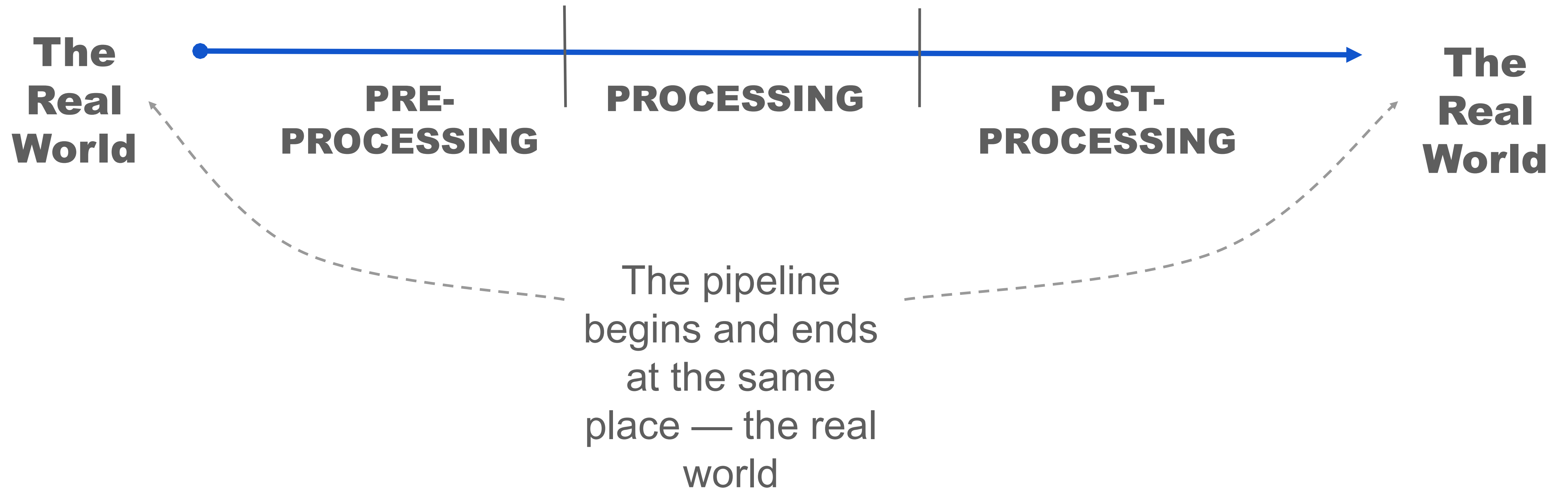
Herb Simon, AI pioneer and CMU faculty
Scientific American September 1995

visualization ***pictures***
The purpose of computing is insight, not numbers.

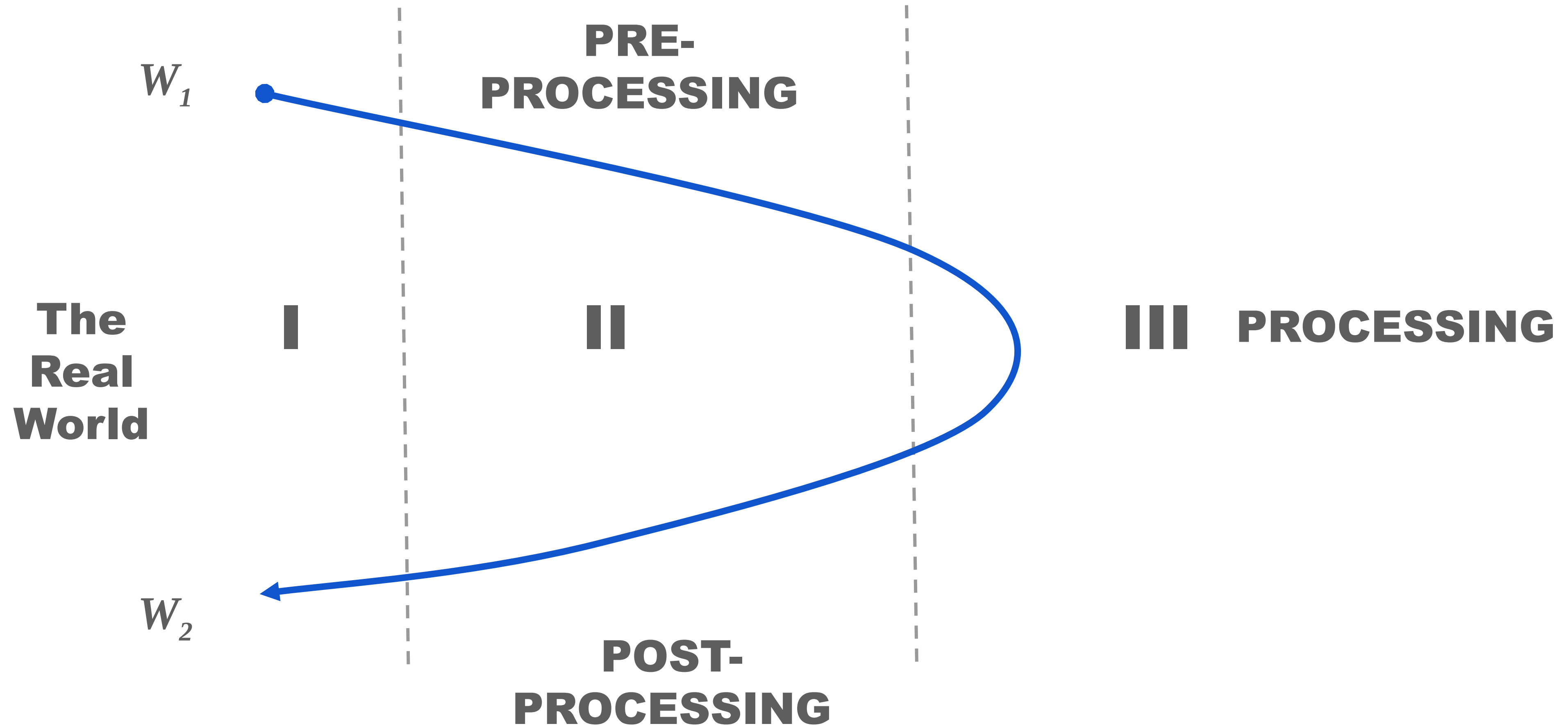
- Richard Wesley Hamming
- Card, Mackinlay, Shneiderman

Why visualize things?

The Data Pipeline



The Pipeline as Arc with Three Zones



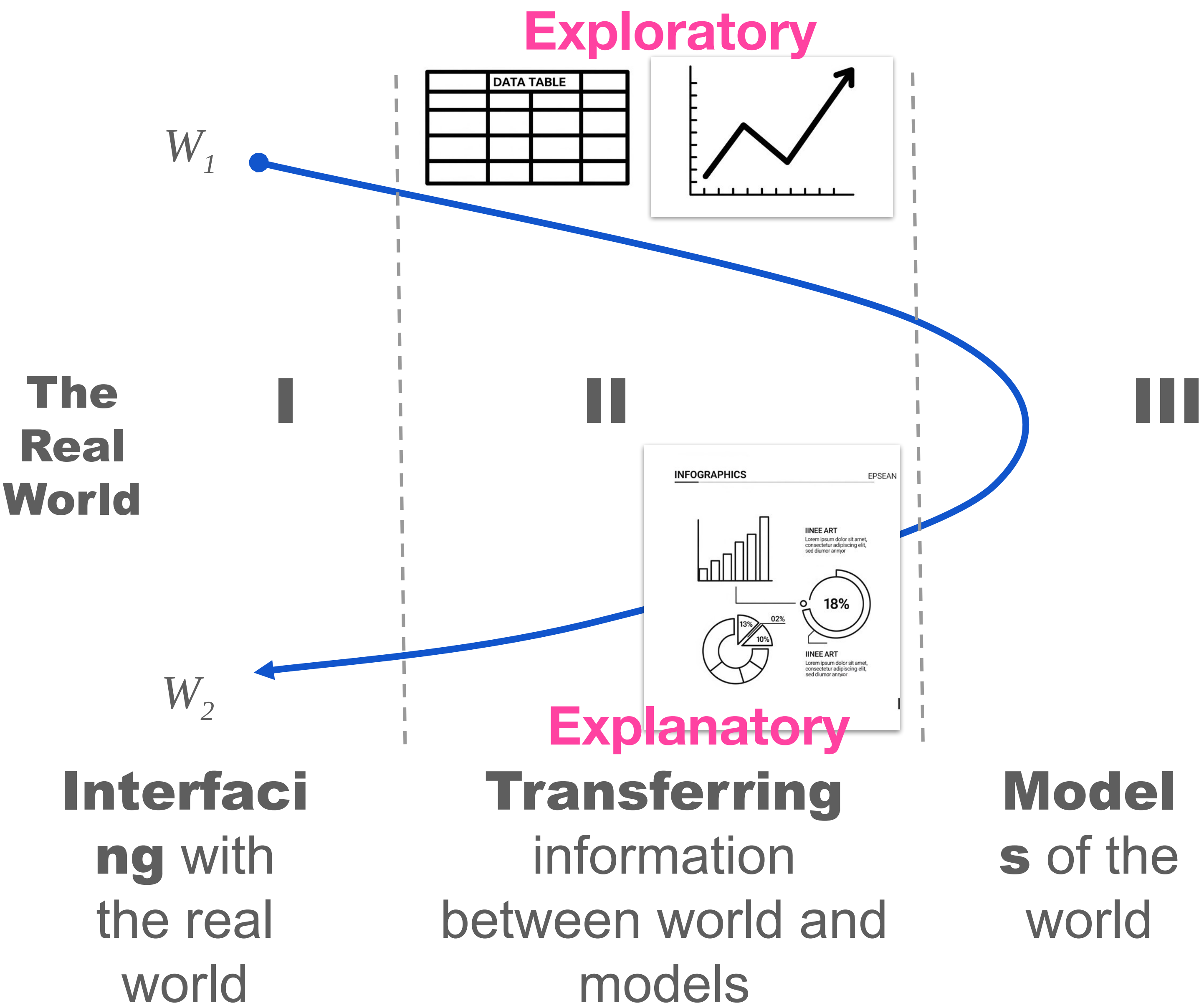
What do the pre- and post-processing phases have in **common**?

Design occurs at the **interface**
between the **world** and
and **models**

$$W_1 \rightarrow M \rightarrow W_2$$

It fundamentally involves the work of **representation**

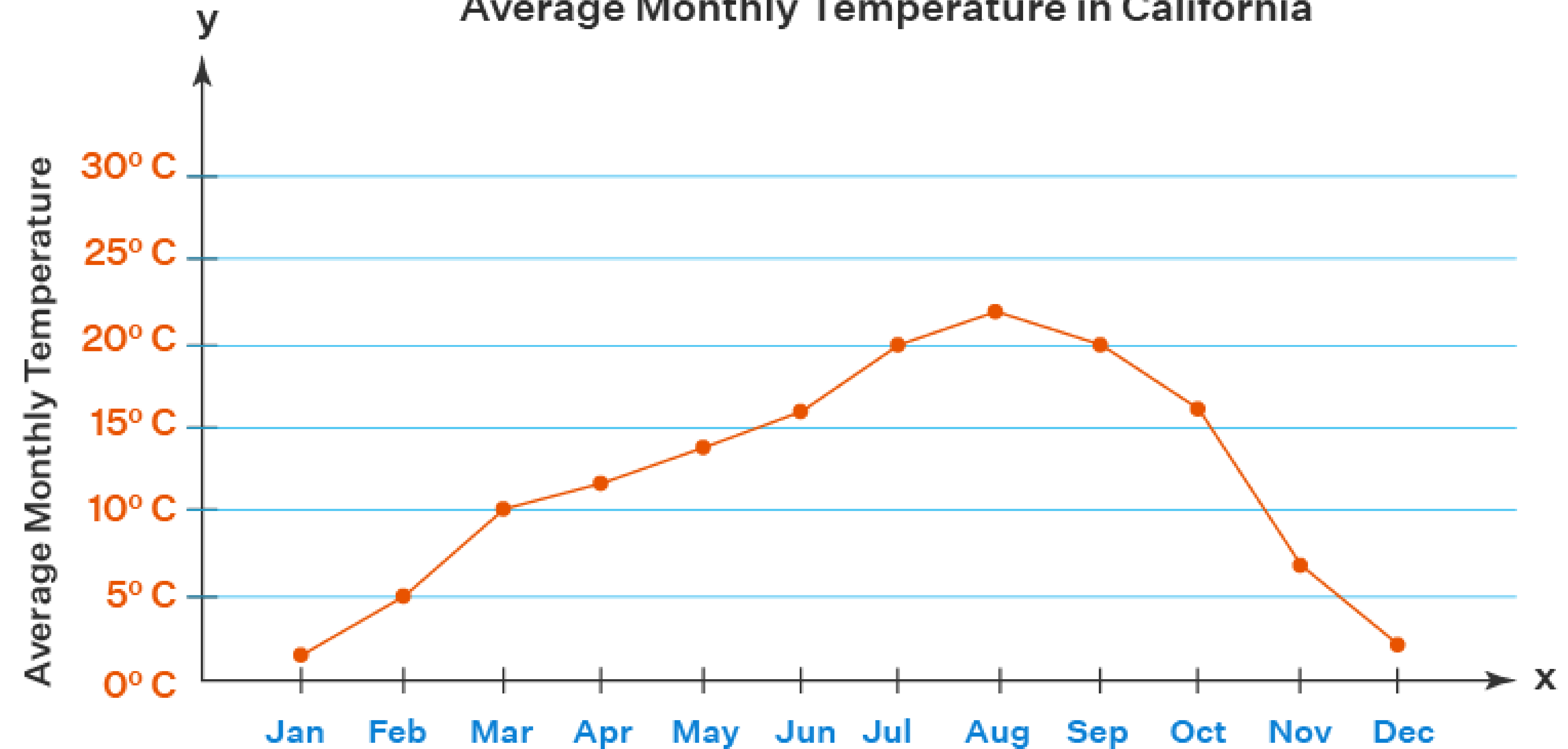
Activities for Each Zone



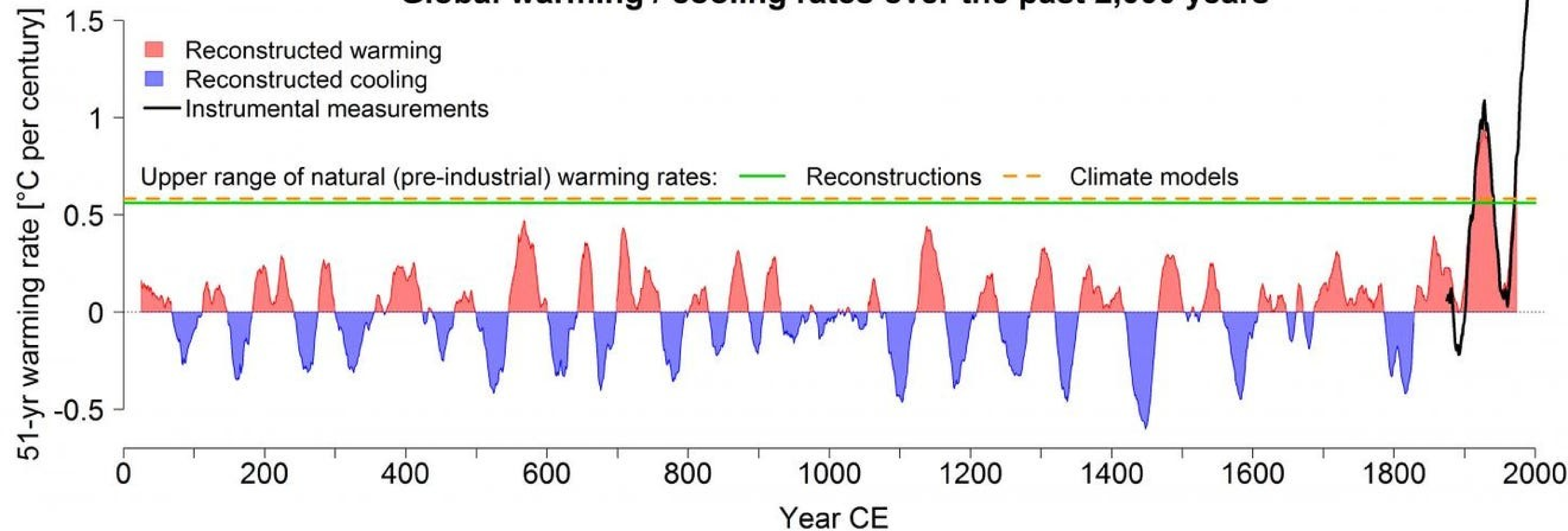
There are many kinds of visualization

Let's look at some examples and identify them as
exploratory or **explanatory**

Average Monthly Temperature in California

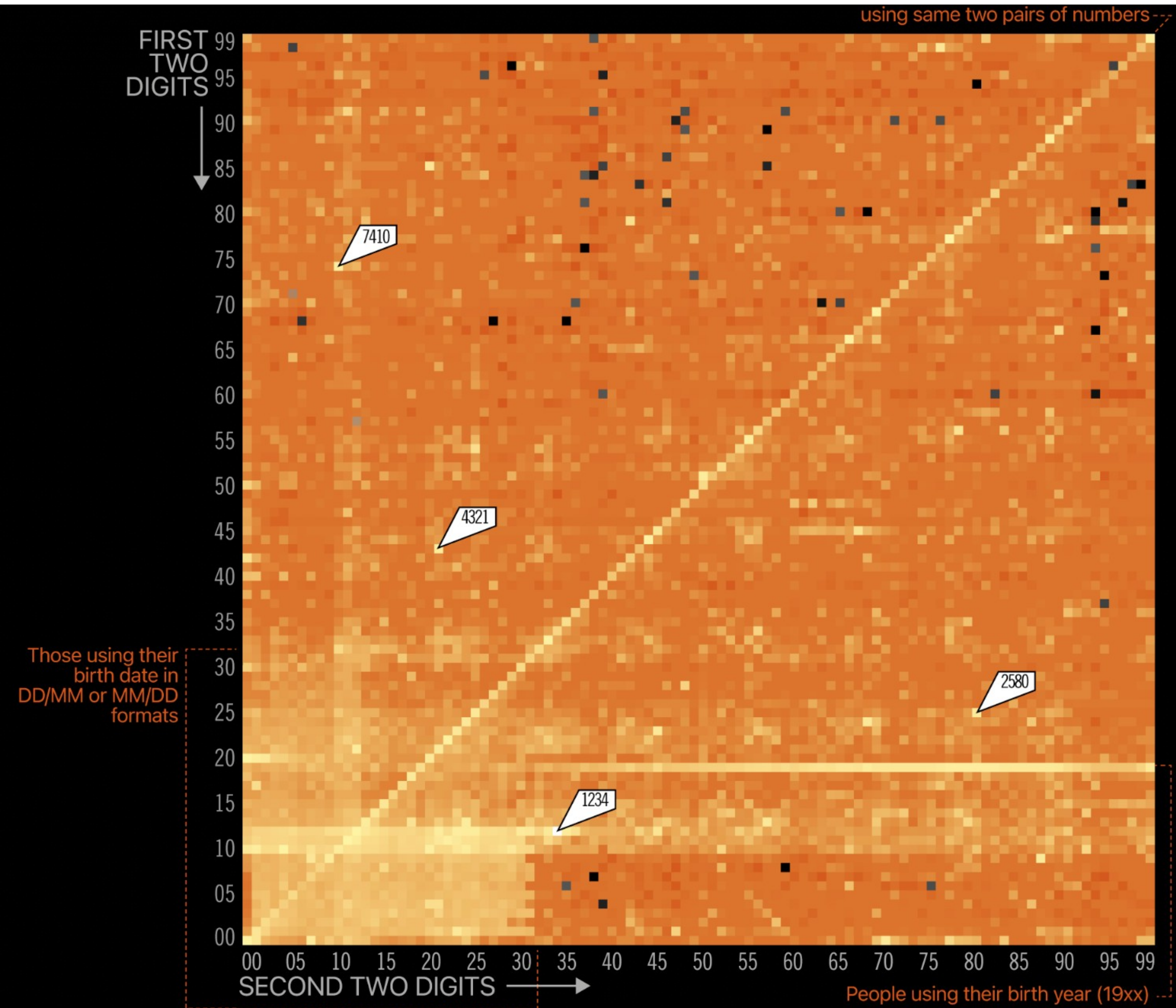


Global warming / cooling rates over the past 2,000 years



Pin Point

The most common 4-digit PIN numbers



most common

1234 0000 7777 2000 2222 9999 5555 1122 8888 2001
 1111 1212 1004 4444 6969 3333 6666 1313 4321 1010

least common

8557 8438 9539 7063 6827 0859 6793 0738 6835 8093
 9047 0439 8196 6093 7394 9480 8398 7637 9629 8068

27% of all PIN numbers

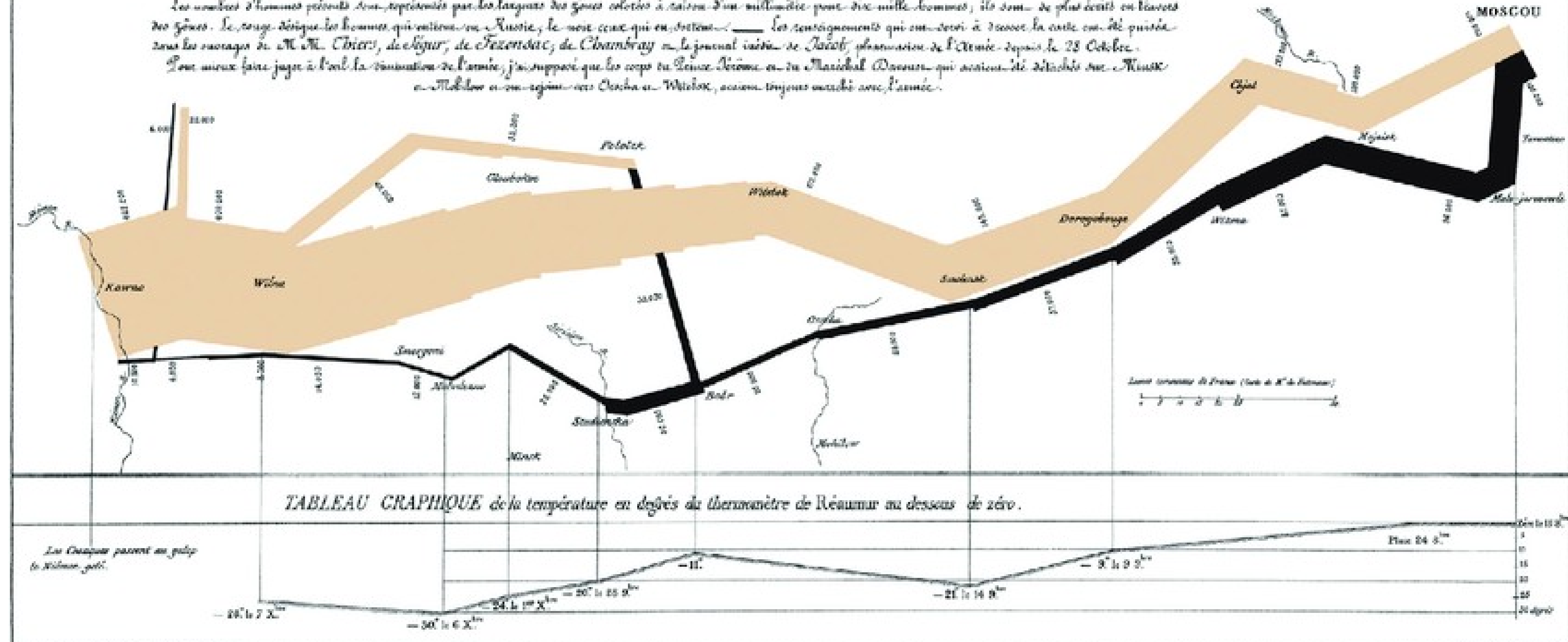
Carte Figurative des pertes successives en hommes de l'Armée Française dans la campagne de Russie 1812-1813.

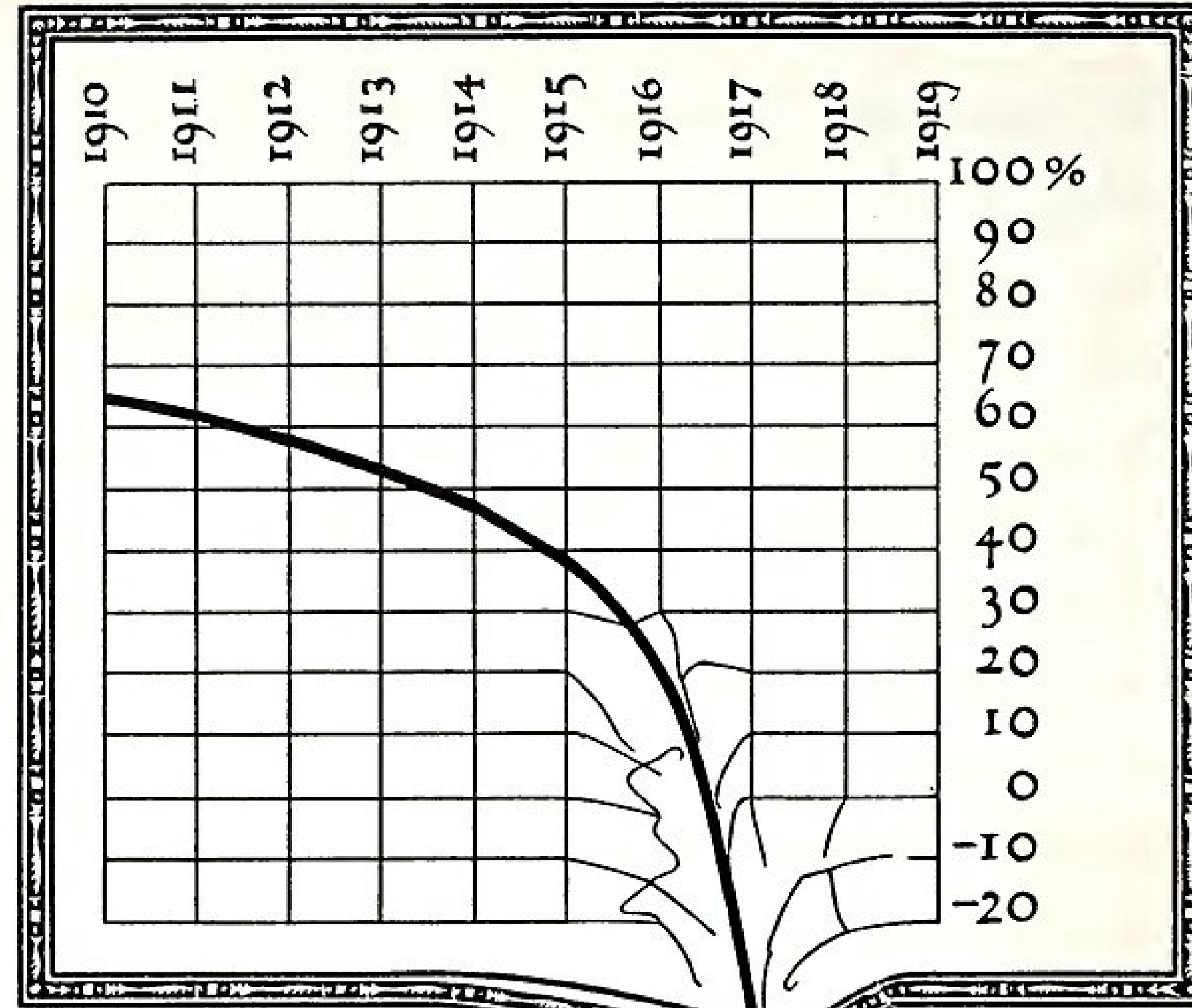
Dessinée par M. Minard, Inspecteur Général des Ponts et Chaussées en retraite.

Paris, le 20 Novembre 1869.

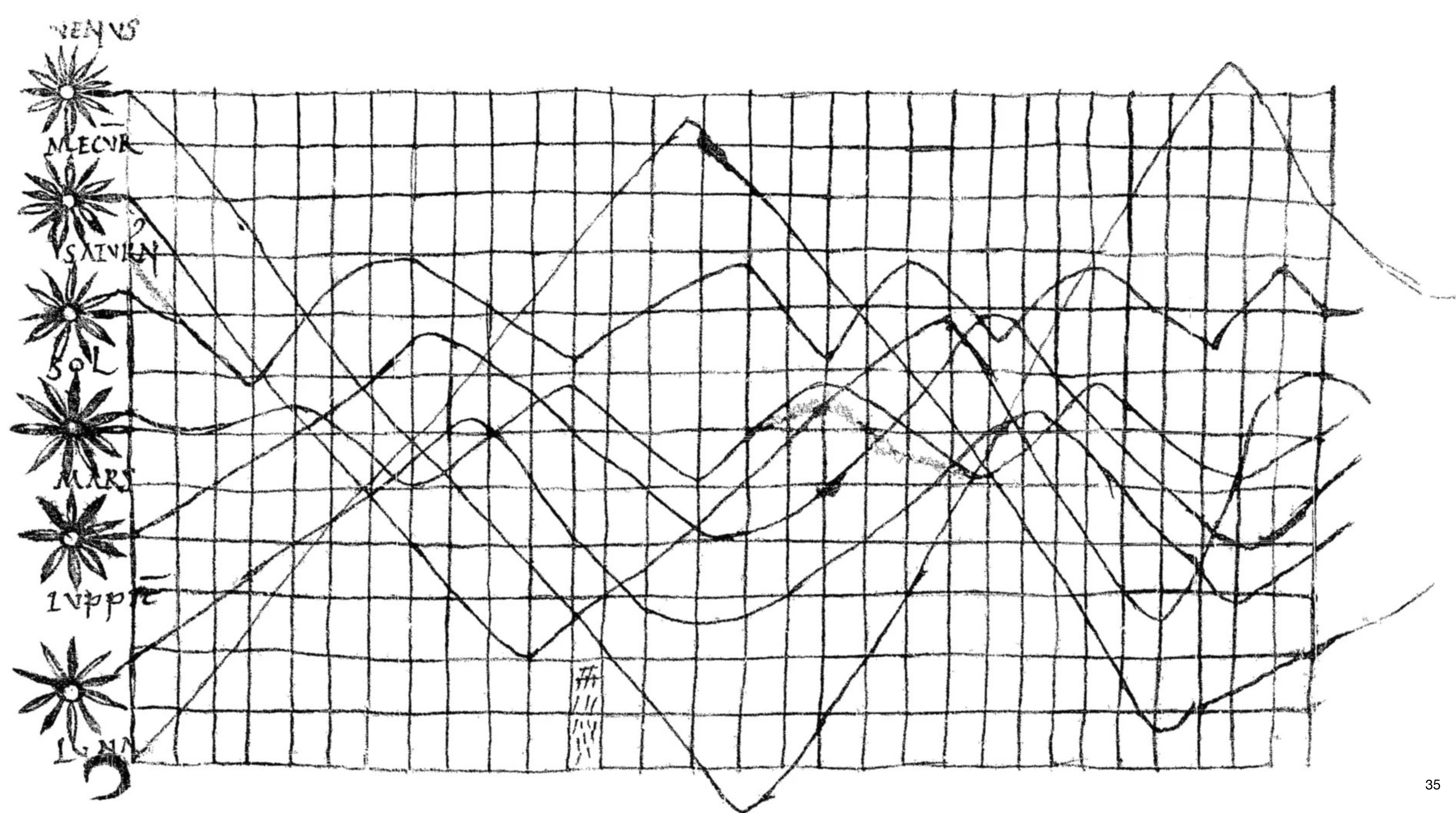
Les nombres d'hommes présents sont représentés par les longueurs des lignes colorées à raison d'un millimètre pour dix mille hommes; ils sont de plus écrits en lettres des lignes. Le rouge désigne les hommes qui entrent en Russie; le noir ceux qui en sortent. — Les renseignements qui ont servi à tracer la carte ont été puisés dans les ouvrages de M. M. Chiers, de Cligny, de Férissac, de Chambray et le journal inédit de Lefebvre, pharmacien de l'Armée depuis le 23 Octobre.

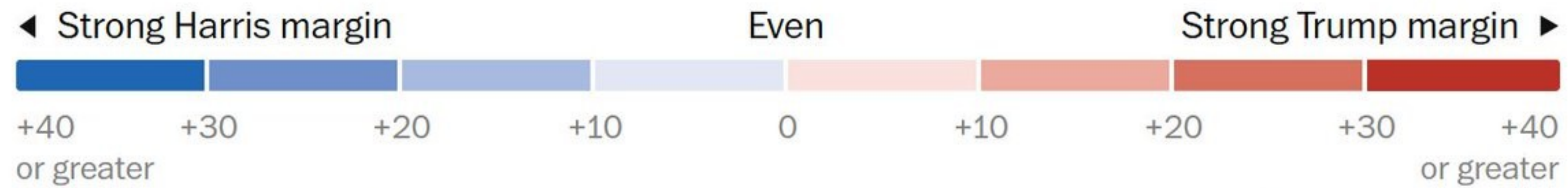
Pour mieux faire juger à l'œil la diminution de l'armée, j'ai supposé que les corps du Prince Jérôme et du Maréchal Dornier qui avaient été détachés sur Minsk et Mohilev n'en reprennent pas l'effectif à Wilna, comme toujours marchés avec l'armée.



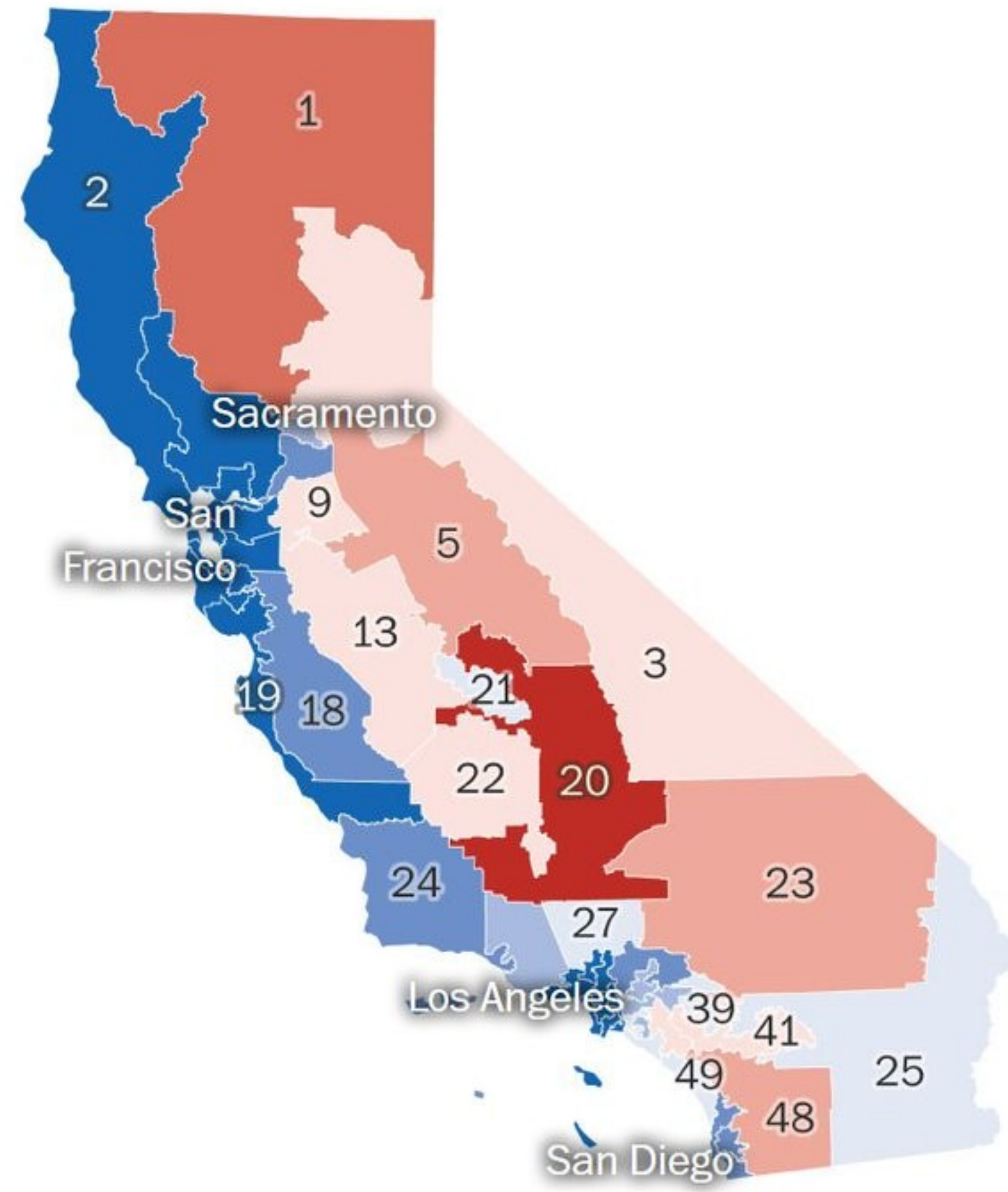


A chart showing the percentage of excellence in the physical properties of books published since 1910.

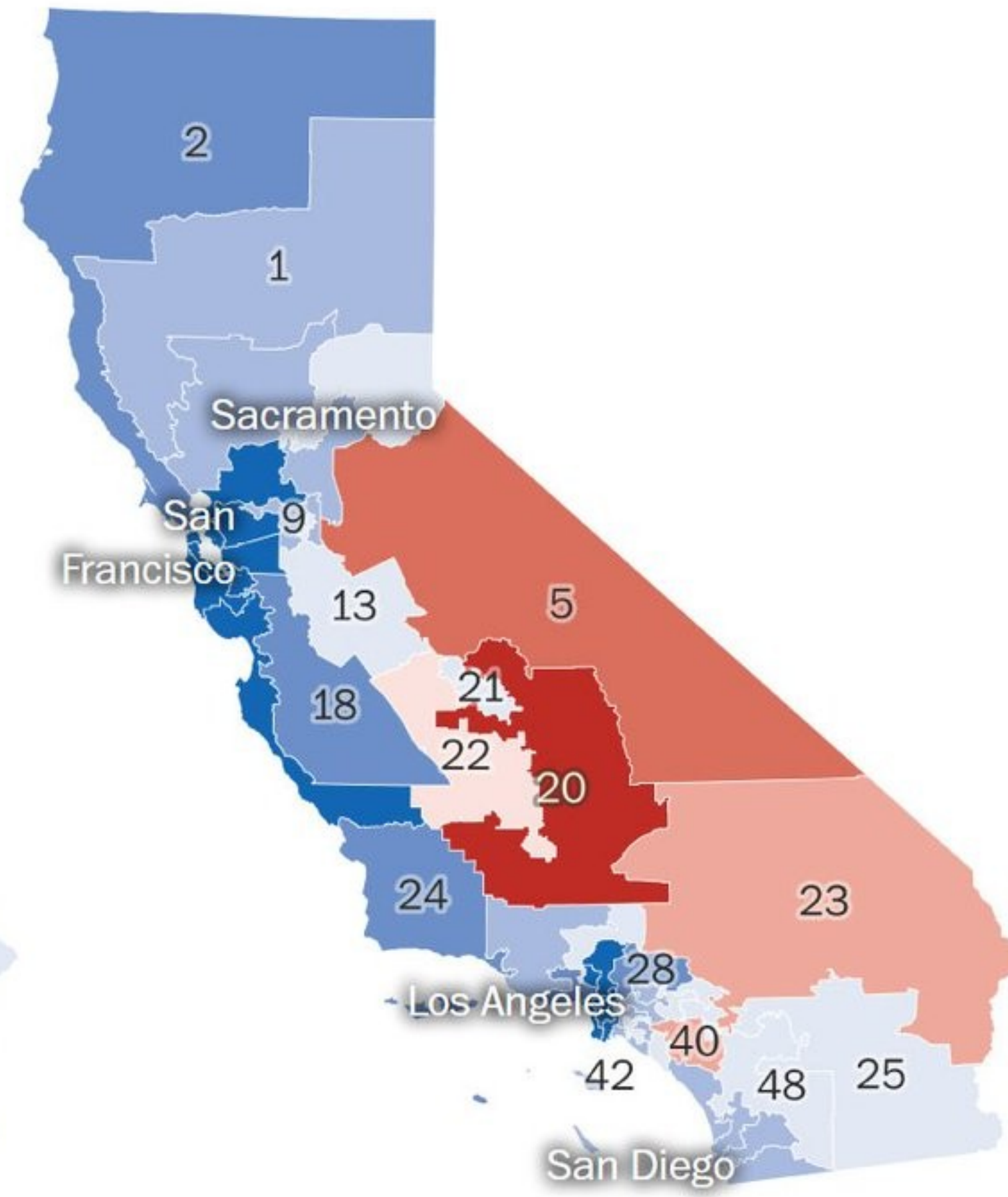




Current map



Proposed map



NICK MOURTOUPALAS/THE WASHINGTON POST

USGS Magnitude 2.5+ Earthquakes, Past Day

33 earthquakes.

☐ Only List Earthquakes Shown on Map

Format

Magnitude

Sort

Newest First

3.9

42 km SSW of Boca de Yuma, ...
2025-08-24 07:41:56 (UTC-04:00) 87.0 km

4.9

164 km ESE of Petropavlovsk...
2025-08-24 07:04:29 (UTC-04:00) 34.1 km

2.6

26 km NNE of Ursine, Nevada
2025-08-24 06:56:20 (UTC-04:00) 5.3 km

4.8

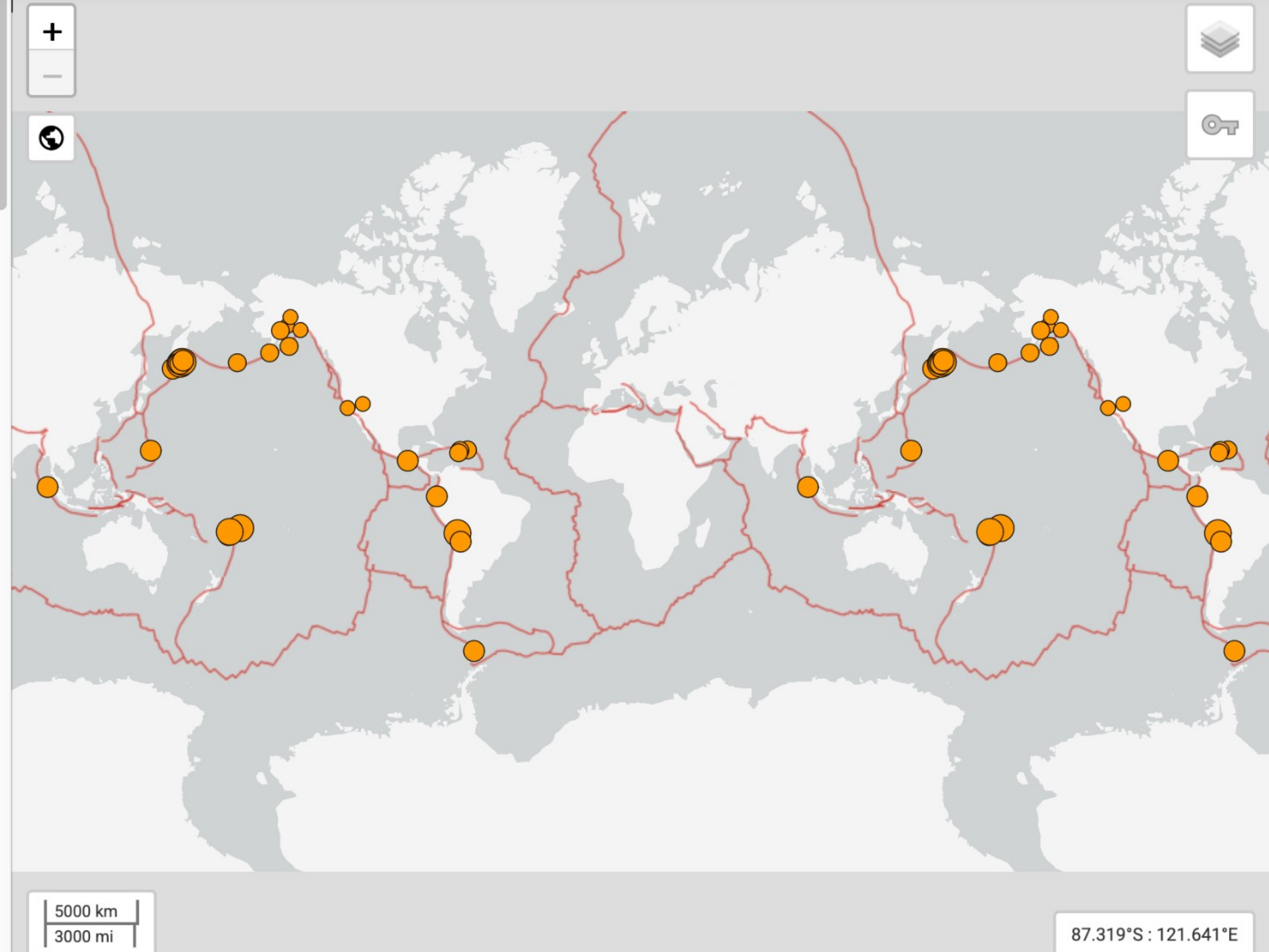
24 km WSW of Alausí, Ecuador
2025-08-24 06:50:33 (UTC-05:00) 101.1 km

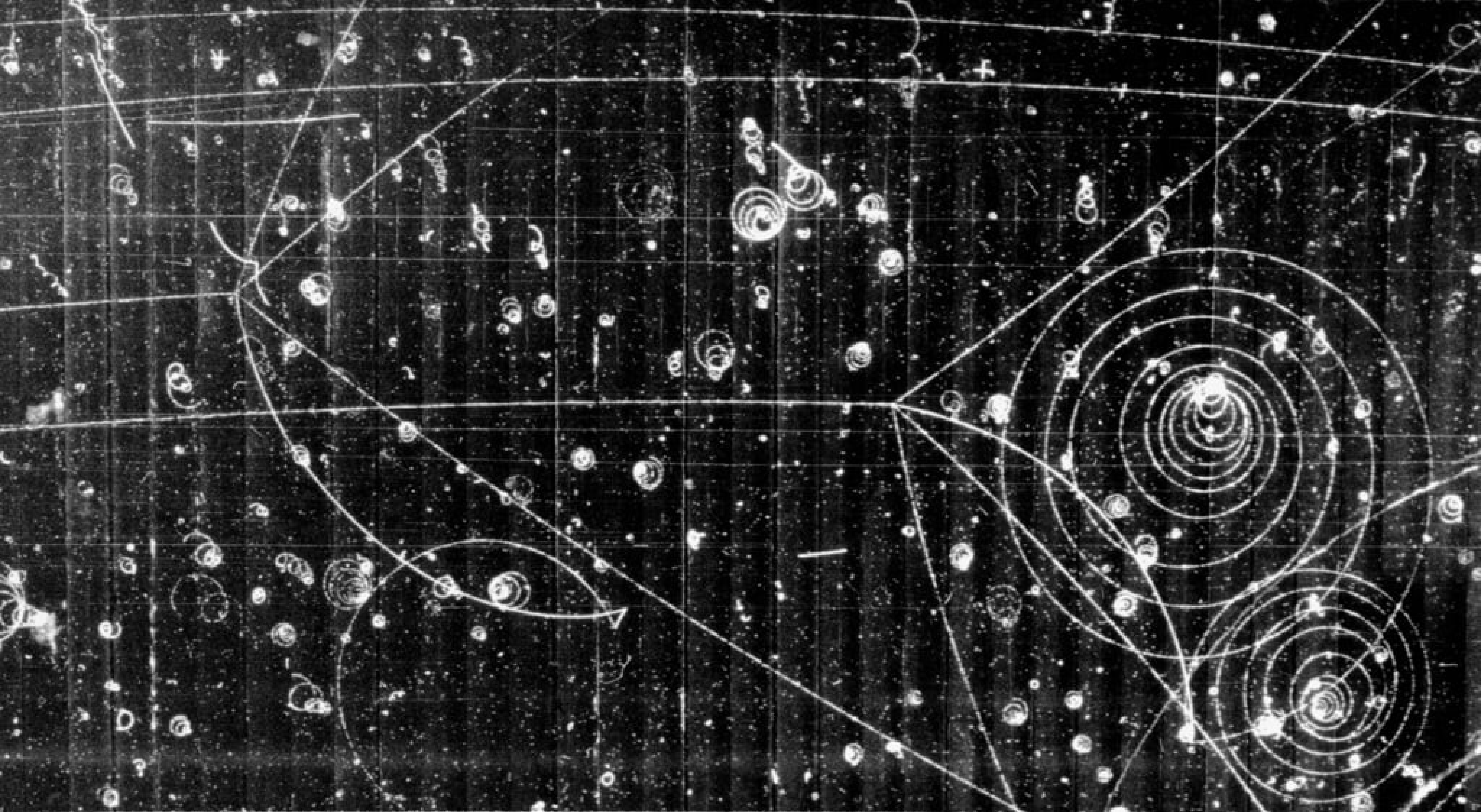
4.9

169 km SSE of Vilyuchinsk, R...
2025-08-24 06:21:53 (UTC-04:00) 38.0 km

3.2

239 km SE of Chiniak, Alaska
2025-08-24 04:30:01 (UTC-04:00) 10.0 km



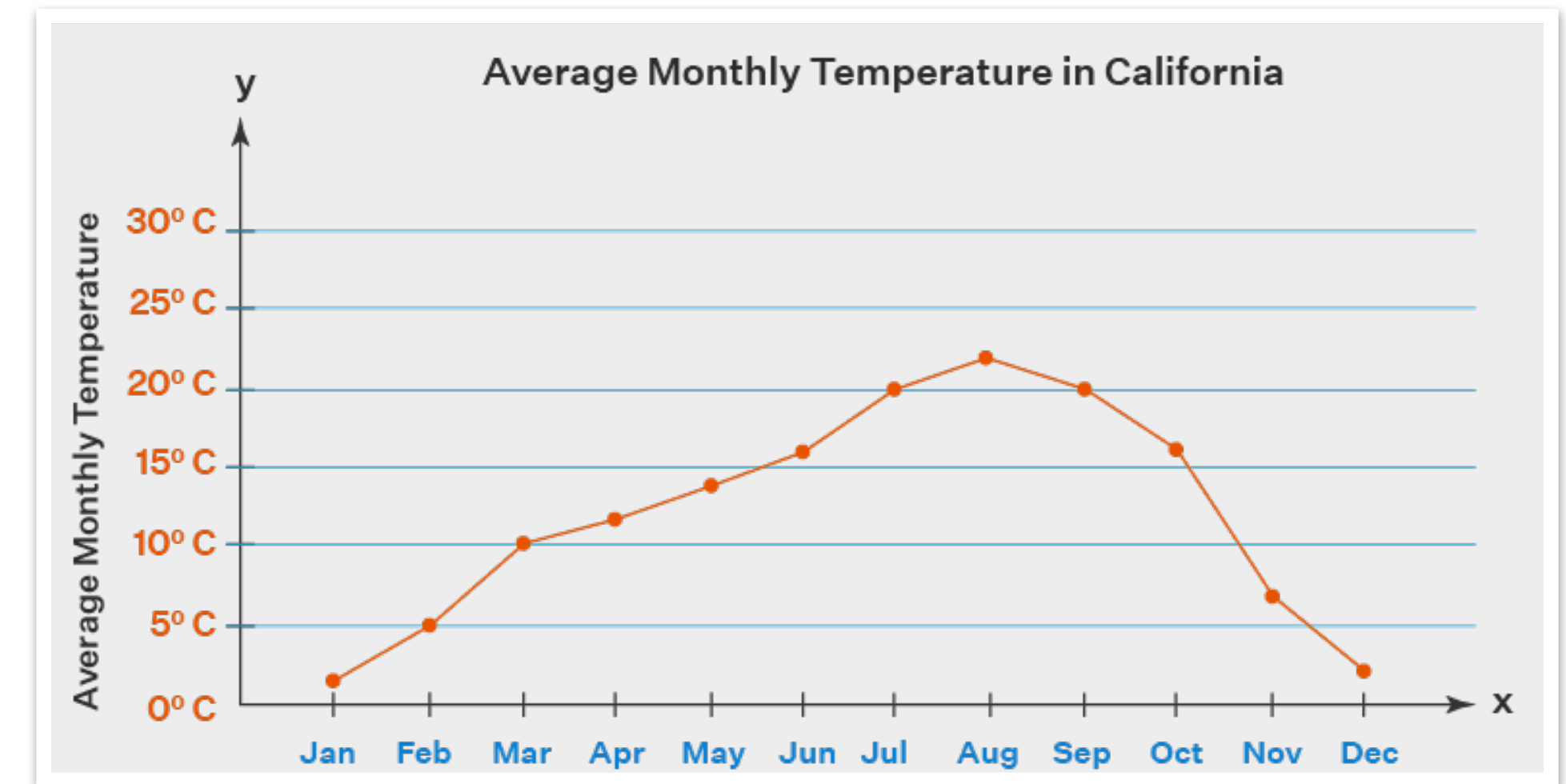


Two Kinds of Visualization

Some graphs are **direct transformations** of the data

These are designed to **describe and explore** data that has not be analyzed

We call these **data visualizations**
exploratory



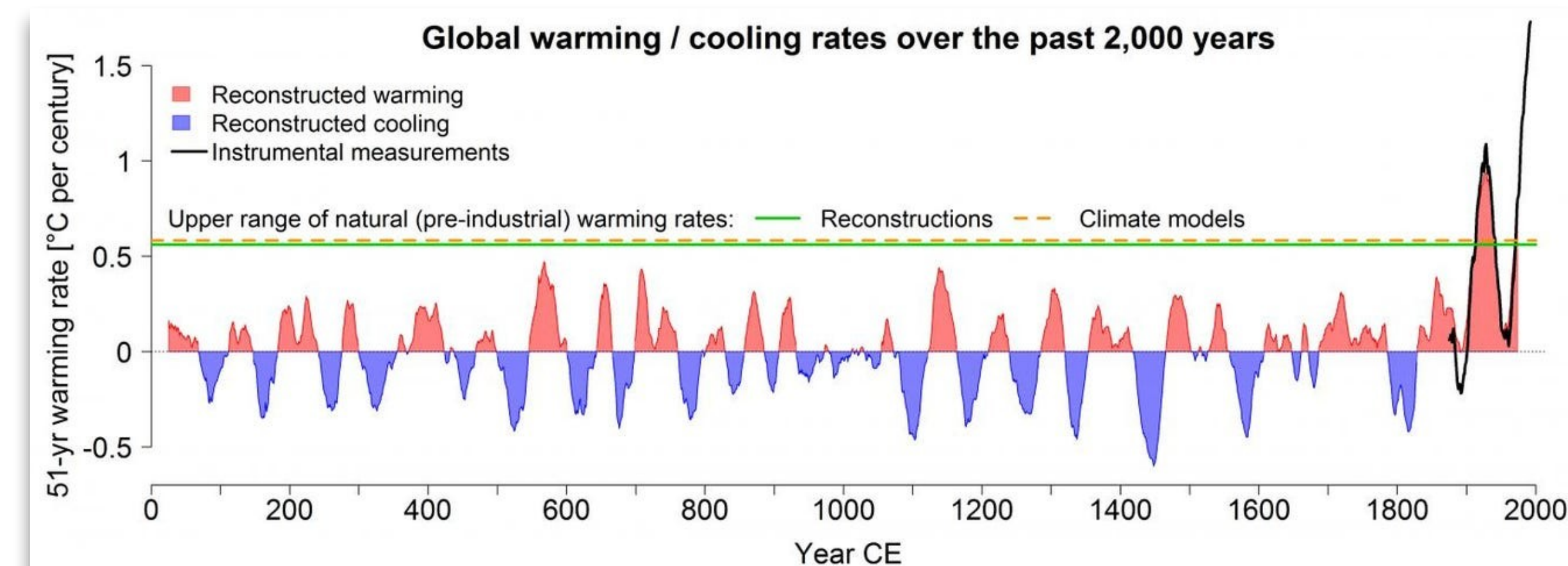
Others are highly **embellished data visualizations**

These are designed to **communicate and explain** results of analysis

These are examples of **storytelling with visualization**

We call these *explanatory*

Infographics, scientific figures, etc.



This **structure of this course** follows this pattern

The **first half** will focus on **exploratory data visualization**

The **second half** will focus on **explanatory storytelling and infographics**

As you leave...

- When you walk around the data science building, notice the different visualizations (and different media!)
- Are they *explanatory* or *exploratory*?
- What is the *intent* of the design? What *insight* do they enable?